- 1 Foundation
  - 1 Introduction
    - 1 Database
      - 1 A database is a structured collection of information or data, usually stored electronically in a computer system.
      - 2 It helps users efficiently store, retrieve, and manage their data
    - 2 Database Management System (DBMS)
      - 1 A Database Management System (DBMS) is software that helps in creating, managing, and maintaining databases.
      - 2 It provides an interface for users to easily store, retrieve, and update data in a systematic and organized way
    - 3 Example
      - 1 Before DBMS
        - **Order Record - Date: September 1, 1975**

          Order ID: 1
          Customer Name: Rajesh Kumar
          Product: Electric Kettle
          Quantity: 2
          Price: ₹800
          Address: 12 MG Road, Mumbai
          Payment Method: Cash
          Delivery Status: Delivered
          Note: Customer prefers delivery before 5 PM.

          Order ID: 2
          Customer Name: Priya Sharma
          Product: Non-Stick Cookware Set
          Quantity: 1
          Price: ₹1500
          Address: B-5, Patel Nagar, Delhi
          Payment: Credit Card
          Delivery Status: Pending

          Order ID: 3
          Product: Yoga Mat
          Name: Kiran Patil
          Amount: ₹1200
          Address: Flat 2A, Gokhale Residency, Pune
          Payment Method: UPI
          Status: Shipped
          Contact Note: Called to confirm address.

          Order ID: 4
          Customer: Anil Mehta
          Product: LED Bulbs
          Cost: ₹800
          Delivery: C-44, Ashok Vihar, Ahmedabad
          Method of Payment: Debit Card
          Status: Processing
          Remark: Urgent delivery requested.

          Order ID: 5
          Customer: Sita Iyer
          Ordered: Stainless Steel Lunch Box Set
          Quantity: 1
          Price: ₹500
          Address: No. 20, Green Avenue, Bangalore
          Payment: Cash on Delivery
          Status: Delivered

          Order ID: 6
          Name: Vikram Singh
          Item: Pressure Cooker
          Quantity: 1
          Price: ₹2000
          Address: 35 Lajpat Nagar, Jaipur
          Payment Method: Net Banking
          Status: Shipped
          Note: Preferred time of delivery : evening.

          Order ID: 7
          Customer: Geeta Nair
          Product: Induction Stove
          Quantity: 1
          Price: ₹3000
          Address: 78, TTK Road, Chennai
          Payment Method: Credit Card
          Status: Processing
          Special Note: Deliver on weekend.

          Order Record - Date: September 3, 1975

          Order ID: 8
          Name: Mohan Lal
          Item: Table Lamp
          Quantity: 3
          Cost: ₹1200
          Address: 56 Rajendra Nagar, Lucknow
          Payment Method: Cash
          Delivery Status: Delivered

          Order ID: 9
          Customer Name: Ritu Verma
          Ordered: Mixer Grinder
          Quantity: 1
          Price: ₹2500
          Address: 90 Sector-45, Chandigarh
          Payment: UPI
          Status: Shipped

          Order ID: 10
          Customer: Amitabh Joshi
          Product: Electric Iron
          Quantity: 1
          Price: ₹700
          Address: 12 Hawa Mahal Road, Jaipur
          Payment Method: Debit Card
          Status: Pending

        - 1 Free-Form Text
        - 2 Inconsistent Structure
        - 3 Difficult to Query
      - 2 After DBMS
        - 1 In 1970, while at IBM, Edgar F. Codd published a landmark paper titled "A Relational Model of Data for Large Shared Data

          Edgar F. Codd

        - 2 In this paper, he proposed a new way to structure data: storing information in tables (or relations) with rows and columns
        - 3

OrderID	CustomerName	ProductName	Quantity	Price (INR)	Address	PaymentMethod	OrderDate	DeliveryStatus	Notes
1	Rajesh Kumar	Electric Kettle	2	800	12 MG Road, Mumbai	Cash	1975-09-01	Delivered	Prefers delivery before 5 PM
2	Priya Sharma	Non-Stick Cookware Set	1	1500	B-5, Patel Nagar, Delhi	Credit Card	1975-09-01	Pending	-
3	Kiran Patil	Yoga Mat	1	1200	Flat 2A, Gokhale Residency, Pune	UPI	1975-09-01	Shipped	Called to confirm address
4	Anil Mehta	LED Bulbs	3	800	C-44, Ashok Vihar, Ahmedabad	Debit Card	1975-09-01	Processing	Urgent delivery requested
5	Sita Iyer	Stainless Steel Lunch Box Set	1	900	No. 20, Green Avenue, Bangalore	Cash on Delivery	1975-09-02	Delivered	-
6	Vikram Singh	Pressure Cooker	1	2000	35 Lajpat Nagar, Jaipur	Net Banking	1975-09-02	Shipped	Prefers evening delivery
7	Geeta Nair	Induction Stove	1	3000	78, TTK Road, Chennai	Credit Card	1975-09-02	Processing	Deliver on weekend
8	Mohan Lal	Table Lamp	3	1200	56 Rajendra Nagar, Lucknow	Cash	1975-09-03	Delivered	-
9	Ritu Verma	Mixer Grinder	1	2500	90 Sector-45, Chandigarh	UPI	1975-09-03	Shipped	-
10	Amitabh Joshi	Electric Iron	1	700	12 Hawa Mahal Road, Jaipur	Debit Card	1975-09-03	Pending	-

          1 Consistency
          2 Easily Queryable
          3 Reduces Errors
          4 Scalability

        - 4

CustomerID	CustomerName	Address	PaymentMethod
1	Rajesh Kumar	12 MG Road, Mumbai	Cash
2	Priya Sharma	B-5, Patel Nagar, Delhi	Credit Card
3	Kiran Patil	Flat 2A, Gokhale Residency, Pune	UPI
4	Anil Mehta	C-44, Ashok Vihar, Ahmedabad	Debit Card
5	Sita Iyer	No. 20, Green Avenue, Bangalore	Cash on Delivery
6	Vikram Singh	35 Lajpat Nagar, Jaipur	Net Banking
7	Geeta Nair	78, TTK Road, Chennai	Credit Card
8	Mohan Lal	56 Rajendra Nagar, Lucknow	Cash
9	Ritu Verma	90 Sector-45, Chandigarh	UPI
10	Amitabh Joshi	12 Hawa Mahal Road, Jaipur	Debit Card

OrderID	CustomerID	OrderDate	DeliveryStatus	Notes
1	1	1975-09-01	Delivered	Prefers delivery before 5 PM
2	2	1975-09-01	Pending	-
3	3	1975-09-01	Shipped	Called to confirm address
4	4	1975-09-01	Processing	Urgent delivery requested
5	5	1975-09-02	Delivered	-
6	6	1975-09-02	Shipped	Prefers evening delivery
7	7	1975-09-02	Processing	Deliver on weekend
8	8	1975-09-03	Delivered	-
9	9	1975-09-03	Shipped	-
10	10	1975-09-03	Pending	-

ProductID	OrderID	ProductName	Quantity	Price (INR)
1	1	Electric Kettle	2	800
2	2	Non-Stick Cookware Set	1	1500
3	3	Yoga Mat	1	1200
4	4	LED Bulbs	3	800
5	5	Stainless Steel Lunch Box Set	1	900
6	6	Pressure Cooker	1	2000
7	7	Induction Stove	1	3000
8	8	Table Lamp	3	1200
9	9	Mixer Grinder	1	2500
10	10	Electric Iron	1	700

          Relational
  - 2 Popular Database Types and Their DBMS Systems
    - 1 Relational Databases (SQL)
      - 1 MySQL
      - 2 PostgreSQL
      - 3 ORACLE DATABASE
      - 4 Microsoft SQL Server
      - 5 MariaDB
      - 6
    - 2 NoSQL Databases
      - 1 Document-Oriented
        - 1 mongoDB
        - 2 CouchDB
        - 3 Amazon DocumentDB
      - 2 Key-Value Stores
        - 1 redis
        - 2 Amazon DynamoDB
      - 3 Column-Family Stores
        - 1 cassandra
        - 2 HBASE
        - 3 Amazon Keyspaces (for Apache Cassandra)
      - 4 Graph Databases
        - 1 neo4j
        - 2 Amazon Neptune
    - 3 In-Memory Databases
      - 1 redis
      - 2 MEMCACHED
      - 3 Amazon ElastiCache
- 2 What is Relational Database?
  - 1 Introduction
    - 1 A relational database stores data in a structured format using rows and columns within tables.
    - 2 In this type of database, multiple tables are linked to each other based on shared information.
    - 3 This connection between tables helps in efficiently organizing data, allowing you to easily combine and retrieve related information across different tables.
  - 2 Common use cases of relational databases
    - 1 Banking Systems
    - 2 School Management Systems
    - 3 Customer Relationship Management (CRM)
    - 4 Inventory Management
    - 5 Healthcare Systems
  - 3 Terminologies
    - 1 Table
      - 1 The fundamental building block of a relational database.
      - 2 A table consists of rows (records) and columns (fields), much like a spreadsheet.
      - 3 Example
        - 1 Customers Table
        - 2 Orders Table
        - 3 Products Table
    - 2 Row (Record)
      - 1 Each row in a table represents a single, unique entry or data point.
      - 2 Example
        - 1 In the Customers Table, a row like (1, Rajesh Kumar, 12 MG Road, Mumbai, Cash) represents one customer.
        - 2 In the Orders Table, a row like (1, 1, 1975-09-01, Delivered, Prefers delivery before 5 PM) represents one order.
    - 3 Column (Field)
      - 1 A column represents an attribute or characteristic of the data stored in a table.
      - 2 Example
        - 1 In the Customers Table, columns include "CustomerID," "CustomerName," "Address," and "PaymentMethod."
        - 2 In the Orders Table, columns include "OrderID," "CustomerID," "OrderDate," etc.
    - 4 Schema
      - 1 The schema defines the overall structure of both the database and each table within it.
      - 2 It outlines how tables are organized, the relationships between them, and the data types for each column.
      - 3 Example
        - 1 in the Customers Table, the schema would specify that
          - 1 "CustomerID" is an integer
          - 2 "CustomerName" is text
          - 3 "Address" is text
          - 4 "PaymentMethod" is text
        - 2 The schema of the Orders Table defines that
          - 1 OrderID" is an integer
          - 2 "OrderDate" is a date
    - 5 Primary Key
      - 1 A column or a set of columns that uniquely identifies each row in a table
      - 2 Example
        - 1 In the Customers Table, "CustomerID" serves as the primary key since it uniquely identifies each customer.
        - 2 In the Orders Table, "OrderID" is the primary key, ensuring each order is distinct
    - 6 Foreign Key
      - 1 A column that creates a link between two tables by referring to the primary key of another table.
      - 2 Example
        - 1 In the Orders Table, "CustomerID" is a foreign key linking to the "CustomerID" in the Customers Table, establishing a relationship between customers and their orders.
        - 2 In the Products Table, "OrderID" is a foreign key linking to the "OrderID" in the Orders Table, connecting products to specific orders.
    - 7 Index
      - 1 A database structure that improves the speed of data retrieval operations on a table.
      - 2 It works like a table of contents in a book, allowing the database to find specific rows faster without scanning the entire table.
      - 3 Example Although not explicitly shown in the image, an index could be created on columns like "CustomerName" in the Customers Table or "OrderDate" in the Orders Table to speed up search operations.
    - 8 SQL (Structured Query Language)
      - 1 The language used to interact with relational databases.
      - 2 It includes commands to
        - 1 Create tables (CREATE)
        - 2 Insert data (INSERT)
        - 3 Update data (UPDATE)
        - 4 Retrieve data (SELECT)
      - 3 Example While not directly shown in the image, you would use SQL commands to interact with these tables. For example, to fetch all delivered orders,

        ```
        SELECT * FROM Orders WHERE DeliveryStatus = 'Delivered';
        ```

        Subtopic
    - 9 Relationship
      - 2 Relationships can be of different types, such as one-to-one, one-to-many, or many-to-many, depending on how data in one table is associated with data in another.
      - 3 Example The Customers Table and Orders Table have a one-to-many relationship, where one customer can have many orders linked through the "CustomerID."
    - 10 Normalization
      - 1 The process of organizing data in a relational database to reduce redundancy and improve data integrity.
      - 2 It involves splitting a table into smaller, related tables and defining relationships between them.
      - 3 Example Instead of storing customer information in the Orders Table, only the "CustomerID" is stored, and customer details are kept in the Customers Table. This separation reduces duplication and ensures consistency.